

UNCOVERING STRATEGY COLLAPSE IN RL-TRAINED LARGE LANGUAGE MODELS

CS639 NLP Project Proposal

Team 10

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PROBLEM DESCRIPTION

- Reinforcement Learning (RL) improves LLM accuracy but may reduce **reasoning diversity** by encouraging models to rely on a small set of high-reward strategies.
- Current evaluation metrics (e.g., **accuracy or pass@k**) cannot explain **how reasoning processes change after RL training**.
- We aim to analyze **reasoning trajectories** to detect whether RL training leads to **strategy collapse**.

SIGNIFICANCE

- Understanding reasoning diversity helps improve **LLM reliability and interpretability**.
- Detecting strategy collapse can inform **better training methods for reasoning models**.
- Our work provides tools for **analyzing reasoning structures** beyond simple accuracy metrics.



TECHNICAL CHALLENGES

- Limited Evaluation Metrics
 - Only evaluation on entire inference
 - Accuracy
 - Pass@1
 - Lack evaluation on details
- Attention Weight
 - Trained to predict the next word rather than inference
 - Led to focus on logically not significant information
- Excess nodes
 - Same meaning, different representation

PROPOSED METHODS OR EXPLORATIONS

01. Rollout

For each question and model, sample N rollouts using temperature > 0 . Parse rollout into steps

02. Construct Reasoning Graph

Cluster similar steps across rollouts into a “state”, then build a branching reasoning graph and compute structure metrics (branching, convergence, path entropy)

03. Intervention

Remove step, resample completions from edit, measure change in final-answer distribution, calculate anchor score

04. Evaluation

Compare models based on reasoning diversity, anchor concentration, and recovery accuracy
Goal: detect strategy collapse (accuracy $\uparrow$, diversity $\downarrow$)

05. Visualization

Generate side-by-side visualization for each model and its respective reasoning graph with toggle option to remove anchor steps and regenerate paths

Novelty

Instead of only showing the model thought process or which step mattered, we also test what happens when we remove the key step. If the RL model can't find another way to solve it, that's clear evidence of strategy collapse.

DATASETS

To analyze reasoning diversity, we use datasets that require multi-step reasoning and allow models to produce chain-of-thought explanations.

- **GSM8K** – A dataset of grade school math word problems designed to test multi-step mathematical reasoning in language models.
- **MATH** – A more challenging dataset of competition-level math problems that require complex reasoning strategies.

EVALUATION METRICS

Accuracy Metrics

- Pass@1 – The probability that the model produces the correct answer in a single attempt.
- Pass@k – The probability that at least one correct answer appears within k sampled outputs.

Reasoning Diversity Metrics

- Strategy Diversity – Measures how many distinct reasoning strategies the model uses.
 $D = \text{no. of reasoning clusters} / \text{total traces}$
- Strategy Entropy – Measures how evenly strategies are distributed.
 $H = - \sum p_i \log p_i$

Graph-Based Metrics

- Branching Factor – Higher branching indicates more exploration of reasoning paths.
 $B = \text{no. of edges} / \text{no. of nodes}$

BASELINE METHODS

We compare reasoning behavior across three model stages:

- **Base Model:** The pretrained model without task-specific fine-tuning.
- **SFT Model:** The model trained using supervised fine-tuning on reasoning datasets.
- **RL Model:** The model further optimized with reinforcement learning for improved accuracy.

For each model, we generate multiple reasoning trajectories and analyze them using reasoning graphs and clustering to compare reasoning diversity.

COMPUTE ESTIMATION

- not computationally expensive, single CPU or GPU
- Generating reasoning trajectories for ~100 problems with 10 samples each
 - 1-2 hours per model, 3 models: 3-6 hours

MODEL CHECKPOINTS

- **Base:** Qwen-2.5-7B
- **SFT:** Qwen-2.5-7B-Instruct
- **RL:** DeepSeek-R1-Distill-Qwen-7B
- All 7B parameter models, same Qwen architecture family, runnable on a single instructional GPU

**THANK
YOU.**

